

Dietary fibre and risk of breast cancer in the UK Women's Cohort Study.

BACKGROUND: Reports of relationships between dietary fibre intake and breast cancer have been inconsistent. Previous cohort studies have been limited by a narrow range of intakes. **METHODS:** Women who developed invasive breast cancer, 350 post-menopausally and 257 pre-menopausally, during 240,959 person-years of follow-up in the UK Women's Cohort Study (UKWCS) were studied. This cohort has 35,792 subjects with a wide range of exposure to dietary fibre with intakes of total fibre in the lowest quintile of <20 g/day up to >30 g/day in the top quintile. Fibre and breast cancer relationships were explored using Cox regression modelling adjusted for measurement error. Effects of fibre, adjusting for confounders were examined for pre- and post-menopausal women separately. **RESULTS:** In pre-menopausal, but not post-menopausal women a statistically significant inverse relationship was found between total fibre intake and risk of breast cancer (P for trend = 0.01). The top quintile of fibre intake was associated with a hazard ratio of 0.48 [95% confidence interval (CI) 0.24-0.96] compared with the lowest quintile. Pre-menopausally, fibre from cereals was inversely associated with risk of breast cancer (P for trend = 0.05) and fibre from fruit had a borderline inverse relationship (P for trend = 0.09). A further model including dietary folate strengthened the significance of the inverse relationship between total fibre and pre-menopausal breast cancer. **CONCLUSIONS:** These findings suggest that in pre-menopausal women, total fibre is protective against breast cancer; in particular, fibre from cereals and possibly fruit.